

① Processos Estocásticos (Dia 21/09/2026)

Martingais (Tempo discreto)

Def: Filtração é $(\mathcal{F}_n)_{n \in \mathbb{N}}$, $\mathcal{F}_n$ σ -álgebra, $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$.

Exemplo: $\mathcal{F}_n = \sigma(X_0, X_1, \dots, X_n)$

Def: Sequência $(M_n)_{n \in \mathbb{N}}$ de v.a.'s é um martingal com respeito à filtração $(\mathcal{F}_n)_{n \in \mathbb{N}}$ se, $\forall n \in \mathbb{N}$:

(i) Adaptado: $M_n \in \mathcal{F}_n$

(ii) Integravel: $\mathbb{E}|M_n| < \infty$

(iii) Propriedade Martingal: $\mathbb{E}(M_n | \mathcal{F}_{n-1}) = M_{n-1}$

②
Obs: • $\mathbb{E}(M_N | \mathcal{F}_n) = M_n \quad \forall N \geq n$

• $\mathbb{E}(M_n) = \mathbb{E}(\mathbb{E}(M_n | \mathcal{F}_{n-1})) = \mathbb{E}(M_{n-1}) = \dots = \mathbb{E}(M_0) \quad \forall n \geq 1.$

Def: $(M_n)_{n \in \mathbb{N}}$ é sub-martingal com respeito a filtração $(\mathcal{F}_n)_{n \in \mathbb{N}}$ se

(i) $M_n \in \mathcal{F}_n$

(ii) $\mathbb{E}|M_n| < \infty$

(iii) $\mathbb{E}(M_n | \mathcal{F}_{n-1}) \geq M_{n-1}$

Prop: $(M_n)_{n \in \mathbb{N}}$ submartingal. $\phi: \mathbb{R} \rightarrow \mathbb{R}$ convexa e crescente.

$\Rightarrow (\phi(M_n))_{n \in \mathbb{N}}$ é sub-martingal.

Se ~~se~~ $(M_n)_{n \in \mathbb{N}}$ é martingal, ϕ não precisa ser crescente pra conclusão.

prova: Jensen $\Rightarrow$

$$\mathbb{E}(\phi(M_n) | \mathcal{F}_{n-1}) \geq \phi(\mathbb{E}(M_n | \mathcal{F}_{n-1})) \geq \phi(M_{n-1}) \quad \forall n \geq 1.$$

Exemplo: $(X_n)_{n \in \mathbb{N}}$ indep., $\mathbb{E}(X_n) = 0$, $\mathbb{E}(S_d) < \infty$.

$$S_n = S_0 + \sum_{i=1}^n X_i \quad \mathcal{F}_n = \sigma(S_0, X_1, \dots, X_n).$$

~~$\mathcal{F}_n = \sigma(S_0, X_1, \dots, X_n)$~~

$\Rightarrow (S_n)_{n \in \mathbb{N}}$ é martingal c.r.a $(\mathcal{F}_n)_{n \in \mathbb{N}}$.

(i) ✓ ~~...~~

~~(i) $\mathbb{E}(S_n) = \sum_{i=1}^n \mathbb{E}(X_i)$ and $\mathbb{E}(S_0) = 0$~~

(ii) $\mathbb{E}|S_n| \leq \mathbb{E}|S_0| + \sum_{i=1}^n \mathbb{E}|X_i| < \infty$. ✓

(iii)
$$\begin{aligned} \mathbb{E}(S_n | \mathcal{F}_{n-1}) &= \sum_{i=1}^n \mathbb{E}(X_i | \mathcal{F}_{n-1}) \\ &= \sum_{i=1}^{n-1} X_i + \underbrace{\mathbb{E}(X_n | \mathcal{F}_{n-1})}_{=\mathbb{E}(X_n)=0} = S_{n-1} + 0. \end{aligned}$$

Exemplo 2: $(X_n)_{n \in \mathbb{N}}$ indep. $\text{Var}(X_n) = \sigma^2 < \infty$.

$$\begin{aligned} \mathbb{E}(S_n^2) &= \mathbb{E}((S_{n-1} + X_n)^2) = \mathbb{E}(S_{n-1}^2) + 2\mathbb{E}(S_{n-1}X_n) + \mathbb{E}(X_n^2) \\ &= \mathbb{E}(S_{n-1}^2) + \sigma^2 = \dots = n\sigma^2 \end{aligned}$$

defina $M_n = S_n^2 - n\sigma^2$. (i) ✓, (ii) ✓

$$\begin{aligned} \mathbb{E}(M_n | \mathcal{F}_{n-1}) &= \mathbb{E}(S_n^2 - n\sigma^2 | \mathcal{F}_{n-1}) \\ &= \mathbb{E}((S_{n-1} + X_n)^2 - n\sigma^2 | \mathcal{F}_{n-1}) \\ &= \mathbb{E}(\underbrace{S_{n-1}^2}_{=M_{n-1} + (n-1)\sigma^2} + \underbrace{2S_{n-1}X_n}_{=0} + \underbrace{X_n^2}_{=\sigma^2} | \mathcal{F}_{n-1}) - \sigma^2 \\ &= M_{n-1} + 0 + \sigma^2 - \sigma^2 = M_{n-1}. \end{aligned}$$

Exemplo 3 $(X_n)_{n \in \mathbb{N}}$, $X_n \geq 0$, $\mathbb{E}(X_n) = 1$.
^{indep.}

$$M_n = \prod_{i=1}^n X_i, \quad M_0 = 1.$$

$$\begin{aligned} \mathbb{E}(M_n | \mathcal{F}_{n-1}) &= \mathbb{E}(X_n M_{n-1} | \mathcal{F}_{n-1}) = M_{n-1} \mathbb{E}(X_n | \mathcal{F}_{n-1}) \\ &= M_{n-1} \mathbb{E}(X_n) = M_{n-1}. \end{aligned}$$

Martingale Transformado

Def: $(A_n)_{n \in \mathbb{N}_+}$ é não antecipativa com respeito a $(\mathcal{F}_n)_{n \in \mathbb{N}}$ se $A_n \in \mathcal{F}_{n-1} \quad \forall n \geq 1$.

Def: Dado $(M_n)_{n \in \mathbb{N}}$, $(A_n)_{n \in \mathbb{N}_+}$,

$$(A \cdot M)_n = M_0 + \sum_{i=1}^n A_i (M_i - M_{i-1})$$

Teo: $(M_n)_{n \in \mathbb{N}}$ martingale c.r.a. $(\mathcal{F}_n)_{n \in \mathbb{N}}$. $(A_n)_{n \in \mathbb{N}_+}$ não antecipativa com respeito a $(\mathcal{F}_n)_{n \in \mathbb{N}}$. Então $((A \cdot M)_n)_{n \in \mathbb{N}}$ é martingale c.r.a. $(\mathcal{F}_n)_{n \in \mathbb{N}}$.
 $|A_n| \leq C$.

prova: (i) ✓

$$(ii) \quad \mathbb{E}(|(A \cdot M)_n|) \leq \mathbb{E}|M_0| + 2C \sum_{i=1}^n \mathbb{E}|M_i| \quad \checkmark$$

$$\begin{aligned} (ii) \quad \mathbb{E}[(A \cdot M)_n - (A \cdot M)_{n-1} | \mathcal{F}_{n-1}] &= \mathbb{E}(A_n (M_n - M_{n-1}) | \mathcal{F}_{n-1}) \\ &= A_n \mathbb{E}(M_n - M_{n-1} | \mathcal{F}_{n-1}) = 0. \end{aligned}$$

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Proposição: $(M_n)_{n \in \mathbb{N}}$ é submartingal, $(A_n)_{n \in \mathbb{N}_+}$ não-antecipativa
 $0 \leq A_i \leq C$. Então $\mathbb{E}((A \cdot M)_n) \leq C \mathbb{E}(M_n) \quad \forall n \in \mathbb{N}$

prova:

$$\mathbb{E}((A \cdot M)_n) = \mathbb{E}\left(\sum_{k=1}^n A_k (M_k - M_{k-1})\right)$$

$$= \sum_{k=1}^n \mathbb{E}(A_k (M_k - M_{k-1}))$$

$$= \sum_{k=1}^n \mathbb{E}\left(\mathbb{E}(A_k (M_k - M_{k-1}) \mid \mathcal{F}_{k-1})\right)$$

$$= \sum_{k=1}^n \mathbb{E}\left(A_k \underbrace{\mathbb{E}(M_k - M_{k-1} \mid \mathcal{F}_{k-1})}_{\geq 0}\right)$$

$$\leq C \sum_{k=1}^n \mathbb{E}(\mathbb{E}(M_k - M_{k-1} \mid \mathcal{F}_{k-1}))$$

$$= C \sum_{k=1}^n \mathbb{E}(M_k - M_{k-1}) = C \mathbb{E}(M_n) \quad \blacksquare$$

Proposição: $(M_n)_{n \in \mathbb{N}}$ submartingal, $(A_n)_{n \in \mathbb{N}}$ não-antecipativa,

$$0 \leq A_n \leq C$$

$$\Rightarrow \mathbb{E}[(A \cdot M)_n] \leq C \mathbb{E}(M_n) \quad \forall n \in \mathbb{N}.$$

prova: exercício $\blacksquare$

⑥ Teorema (Decomposição de Doob): $(Y_n)_{n \in \mathbb{N}}$ submartingal
 $\Rightarrow \exists$ martingal $(M_n)_{n \in \mathbb{N}}$, $\exists (A_n)_{n \in \mathbb{N}}$ não antecipativa, $A_0 = 0$ t.q

$$Y_n = M_n + A_n$$

Prova: Exercício. ~~11~~

Tempo de Parada Opcional

Def: v.a. τ em $\mathbb{N} \cup \{+\infty\}$ é tempo de parada c.r. a $(\mathcal{F}_n)_{n \in \mathbb{N}}$ se
 $\{\tau \leq n\} \in \mathcal{F}_n$, $\forall n \in \mathbb{N}$.

Def: ~~###~~ ^{Dado} $\tau < \infty$ q.c., $Y_\tau := \sum_{k=0}^{\infty} \mathbb{1}_{\{\tau \geq k\}} Y_k$ (~~###~~ $Y_{\tau(\omega)}(\omega) = Y_\tau(\omega)$)

Obs: tempo de parada $\tau \Rightarrow n \wedge \tau = \min\{n, \tau\}$ é tempo de parada finito

$(Y_{n \wedge \tau})_{n \in \mathbb{N}}$: "processo parado"

Teo: $(M_n)_{n \in \mathbb{N}}$ martingal c.r. a $(\mathcal{F}_n)_{n \in \mathbb{N}}$
 τ tempo de parada c.r. a $(\mathcal{F}_n)_{n \in \mathbb{N}}$

~~###~~
 $\Rightarrow (M_{n \wedge \tau})_{n \in \mathbb{N}}$ é martingal c.r. a $(\mathcal{F}_n)_{n \in \mathbb{N}}$

prova: exercício ~~11~~

7) Obs: ~~Ainda não temos~~ Ainda não temos $\mathbb{E}(M_\tau) = \mathbb{E}(M_0)$.

Teo (§ 1º Teorema da Parada Opcional)

$(M_n)_{n \in \mathbb{N}}$ martingal, τ tempo de parada finito,

$(M_{n \wedge \tau})_{n \in \mathbb{N}}$ limitado (isto é $|M_{n \wedge \tau}| \leq C \forall n$). Então

$$\mathbb{E}(M_\tau) = \mathbb{E}(M_0)$$

prova: $\tau < +\infty \Rightarrow M_{\tau \wedge n} \xrightarrow{n \rightarrow \infty} M_\tau \text{ q.c.}$

$$|M_{\tau \wedge n}| \leq C \quad \forall n \in \mathbb{N} \xRightarrow{\text{TCD}} \mathbb{E}(M_\tau) = \lim_{n \rightarrow \infty} \mathbb{E}(M_{\tau \wedge n}) = \mathbb{E}(M_0)$$

Teo (Parada Opcional de Doob): $(M_n)_{n \in \mathbb{N}}$ martingal, τ tempo de parada

$\Rightarrow \mathbb{E}(M_\tau) = \mathbb{E}(M_0)$ se valer algum item abaixo:

(i) τ é limitado ($|\tau| \leq C \text{ q.c.}$)

(ii) $(M_{\tau \wedge n})_{n \in \mathbb{N}}$ é limitado e $\tau < \infty \text{ q.c.}$

(iii) $\mathbb{E}(\tau) < \infty$, $|M_k - M_{k-1}| \leq C$, pra algum $C > 0$

(iv) $\mathbb{E}(\tau) < \infty$ e $\mathbb{E}(|M_k - M_{k-1}| | \mathcal{F}_{k-1}) \leq C$, pra algum $C > 0$.

prova: (i) $\checkmark$, (ii) $\checkmark$

$$(iii) \quad M_{\tau \wedge n} = M_0 + \sum_{k=1}^n \mathbb{1}_{\{\tau \geq k\}} (M_k - M_{k-1})$$

$$\tau < \infty \Rightarrow M_{\tau \wedge n} \xrightarrow{n \rightarrow \infty} M_\tau \text{ q.c. } (*)$$